Exercise 4.1.1. Let T be left multiplication by the matrix $\begin{bmatrix} 1 & 2 & 0 & -1 & 5 \\ 2 & 0 & 2 & 0 & 1 \\ 1 & 1 & -1 & 3 & 2 \\ 0 & 3 & -3 & 2 & 6 \end{bmatrix}$. Compute ker$T$ and im$T$ explicitly by exhibiting bases for these spaces, and verify (1.7).

Solution. One basis of ker$T$ can be $\{(-7,5,7,3,0)', (0,-5,-1,0,2)'\}$ and one basis of im$T$ can be $\{(1,2,1,0)', (0,1,2,1)', (0,0,1,1)'\}$. Since $5 = 2+3$ thus satisfies (1.7).

Exercise 4.1.5. Let A be a $k \times m$ matrix and let B be an $n \times p$ matrix. Prove that the rule $M \mapsto AMB$ defines a linear transformation from the space $F^{m\times n}$ of $m \times n$ matrices to the space $F^{k\times p}$.

Proof. We only need to check the homomorphism in addition and scaling multiplicity, which are apparently true.

□

Exercise 4.1.8. Prove that every $m \times n$ matrix A of rank 1 has the form $A = XY^{\mathrm{t}}$, where X, Y are $m-$ and $n-$dimensional column vectors.

Proof. Since the rank of A is 1, there exists one non-zero column i of A such that all other columns can be represented in the form cA_i, say $(c_1, c_2, \ldots, c_n)$. Let $X = A_i$ and $Y = (c_1, c_2, \ldots, c_n)'$ satisfying the condition.

□

Exercise 4.2.2. Find all linear transformations $T : \mathbb{R}^2 \to \mathbb{R}^2$ which carry the line $y = x$ to the line $y = 3x$.

Solution. We know $T(1,1)' = (1,3)'$ which is also sufficient. Thus, we need $T_{11} + T_{12} = \frac{1}{3}(T_{21} + T_{22})$.

Exercise 4.2.5. Let A be an $n \times n$ matrix, and let $V = F^n$ denote the space of row vectors. What is the matrix of the linear operator "right multiplication by A" with respect to the standard basis of V?

Solution. The answer is A^{t}.